C# Machine Problem: Cross-Border FinTech Remittance Processing & AML Compliance Audit Terminal

Problem Scenario

You are engaged as a software engineer by an international financial technology (FinTech) corporation that manages an automated cross-border remittance gateway. The gateway processes inbound foreign currency wire transfers, verifies sender identity records, converts foreign currencies to Philippine Pesos (PHP), screens transactions against strict Anti-Money Laundering (AML) regulatory rules, calculates tiered service commissions, and produces audited daily settlement manifests.

Your task is to build a robust, modular, multi-file console application that demonstrates industrial software craftsmanship:

- Decomposing financial formulas and compliance screening into standalone **modular static methods returning typed C# tuples**.
- Safeguarding the system against crashes using **defensive parsing (`TryParse`)** and array boundary guards.
- Managing multi-record batch manifests dynamically using **generic `List<T>` collections**.
- Executing an **interactive, loop-driven console dispatch terminal** with loop control keywords (`break` , `continue`).
- Reading batch records from disk (`remittance_batch_input.txt`) and exporting audited compliance manifests (`remittance_manifest_output.txt`) using **deterministic resource disposal (`using`)**.
- Organizing the codebase logically across segregated namespaces (`FinTechRemittance.Core` and `FinTechRemittance.IO`).

Learning Competencies Assessed

Competency	Application in FinTech Machine Problem
Defensive Ingestion	Token boundary checking (`parts.Length >= 10`), defensive parsing (`decimal.TryParse` , `int.TryParse` , `bool.TryParse`) with `out` parameters, skipping corrupt records via `continue` .
Loop Control	Continuous interactive terminal execution (`do-while` / `while`), batch iteration (`for` / `foreach`), and clean session termination (`break`).
Collections Management	Synchronizing parallel in-memory records using generic lists (`List<string>` , `List<decimal>` , `List<int>`) without fixed-size array constraints.
Modular Methods & Tuples	Refactoring currency exchange, AML screening, and fee structures into pure static methods returning multi-value named C# tuples.
Namespace Architecture	Strict separation of financial algorithms (`FinTechRemittance.Core`) from file streams and console output (`FinTechRemittance.IO`).
Safe File I/O	Reading CSV records and writing formatted manifests using `StreamWriter` enclosed within `using` statements to guarantee resource release.

Namespace Architecture

Segregate your project code into two clean namespaces:

- `FinTechRemittance.Core` : Houses currency exchange engines, AML compliance evaluators, fee calculators, and batch summary analytics.
- `FinTechRemittance.IO` : Houses batch CSV file ingestion, audited manifest disk export, and tabular console report generators.

Required Modular Methods in `FinTechRemittance.Core.RemittanceEngine`

Refactor all operational algorithms out of top-level code into standalone, testable static methods:

Method A: Currency Conversion & FX Spread Engine

- **Identifier:** `ConvertCurrencyToPhp`
- **Namespace:** `FinTechRemittance.Core`
- **Signature:** `public static (decimal exchangeRate, decimal grossPhpAmount, decimal fxSpreadFee) ConvertCurrencyToPhp(decimal sourceAmount, string currencyCode)`
- **Exchange Rate Table:** USD = ₱58.20 | EUR = ₱62.50 | JPY = ₱0.38 | CAD = ₱42.10 (Unknown currencies evaluate to ₱0.00).
- **Logic:** Multiplies source foreign amount by exchange rate to obtain Gross PHP Amount. Computes institutional FX spread markup (Gross PHP Amount × 1.25%). Returns a named tuple containing rate, gross PHP amount, and FX fee.

Method B: AML Compliance & Sanctions Screening Evaluator

- **Identifier:** `EvaluateAmlStatus`
- **Namespace:** `FinTechRemittance.Core`
- **Signature:** `public static string EvaluateAmlStatus(decimal grossPhpAmount, int corridorCode, string transferPurpose, bool isKycVerified, bool isSanctionedEntity, int senderRiskScore)`
- **Return Values:** `"SETTLEMENT REJECTED"`, `"HELD FOR COMPLIANCE REVIEW"`, or `"APPROVED FOR SETTLEMENT"`
- **Strict Priority Decision Hierarchy:**
  - **Immediate Rejection (Priority 1):** Sanctioned entity flag is true (`isSanctionedEntity == true`), OR gross amount exceeds the statutory ceiling of ₱1,000,000.00, OR sender risk score exceeds 85 (score scale: 1–100), OR corridor code is invalid (< 1 or > 3). Returns `"SETTLEMENT REJECTED"`.
  - **Hold for Manual Audit (Priority 2):** Unverified KYC (`isKycVerified == false`) on amounts > ₱20,000.00, OR high-risk corridor (`corridorCode == 3`) on amounts > ₱50,000.00, OR high-risk purpose (`"Cryptocurrency Arbitrage"` or `"Offshore Real Estate"`), OR sender risk score between 60 and 85 inclusive. Returns `"HELD FOR COMPLIANCE REVIEW"`.
  - **Settlement Cleared (Default):** Returns `"APPROVED FOR SETTLEMENT"`.

Method C: Service Commission & Net Payout Calculator

- **Identifier:** `CalculateRemittanceFees`
- **Namespace:** `FinTechRemittance.Core`
- **Signature:** `public static (decimal baseCommission, decimal channelFee, decimal expressSurcharge, decimal totalFeeDeducted, decimal netPayoutPhp) CalculateRemittanceFees(decimal grossPhpAmount, string deliveryChannel, bool isExpressTransfer, string amlStatus)`
- **Logic:**
  - **Cancelled Condition:** If `amlStatus == "SETTLEMENT REJECTED"`, all fees and net payout evaluate to ₱0.00.
  - **Base Commission (Tiered):** Up to ₱10,000.00: flat ₱150.00. Between ₱10,000.01 and ₱100,000.00: 1.5% of gross PHP. Exceeding ₱100,000.00: 1.0% of gross PHP.
  - **Delivery Channel Fee:** `"Bank Deposit"`: ₱50.00 | `"Cash Pickup"`: ₱120.00 | `"Mobile Wallet"` (GCash/Maya): ₱30.00 | Other: ₱100.00.
  - **Express Surcharge:** If `isExpressTransfer == true`, add ₱250.00 priority routing fee; otherwise ₱0.00.
  - **Total Fee:** $ ext{Base Commission} + ext{Channel Fee} + ext{Express Surcharge}$.
  - **Net Payout:** $ ext{Gross PHP Amount} - ext{Total Fee Deducted}$.

Method D: Batch Settlement Analytics Generator

- **Identifier:** `GenerateSettlementSummary`
- **Namespace:** `FinTechRemittance.Core`
- **Signature:** `public static (int totalTransactions, int approvedCount, int heldCount, int rejectedCount, decimal totalVolumePhp, decimal totalRevenueEarned, decimal avgTransferPhp) GenerateSettlementSummary(List<decimal> grossAmounts, List<decimal> totalFees, List<string> amlStatuses)`
- **Logic:** Iterates across parallel list indices. Tallies counts per status category, aggregates total gross PHP volume (only for approved and held transfers), and calculates total fee revenue collected. Computes average transfer amount guarded with `if (totalTransactions > 0)` to eliminate `DivideByZeroException` on empty batches.

Interactive Console Menu & Operational Flow

The application entry point must run inside a continuous loop (`do-while` or `while`) presenting the dispatch menu:

```
============================================================
     FINTECH CROSS-BORDER REMITTANCE & AML GATEWAY (M2)
============================================================
  1. Process Single Wire Transfer (Interactive Prompt)
  2. Load Batch Transfers from File (remittance_batch_input.txt)
  3. View Active Settlement Ledger & Compliance Analytics
  4. Export Audited Manifest to File (remittance_manifest_output.txt)
  5. Exit Gateway Terminal
============================================================
Enter option (1-5):
```

Menu Option Specifications:

- **Option 1 (Process Single Wire Transfer):** Interactively prompts the operator for sender name, currency (USD/EUR/JPY/CAD), foreign amount, destination corridor (1–3), transfer purpose, KYC verification status (y/n), sanctions status (y/n), risk score (1–100), delivery channel (1–3), and express priority (y/n). Invokes calculation methods, assigns an auto-incremented tracking identifier (e.g., `TXN-3001`), and appends records into active memory collections.
- **Option 2 (Load Batch Transfers from File):** Reads `remittance_batch_input.txt` from disk using `File.ReadAllLines` or `StreamReader` enclosed in a `using` block. Validates record health (`parts.Length >= 10`) and uses `continue` to skip corrupted lines cleanly. Evaluates valid transfers and adds them to active memory.
- **Option 3 (View Active Settlement Ledger & Analytics):** Iterates through stored transfer records using `for` or `foreach` loops, printing a tabular report followed by the computed batch analytics.
- **Option 4 (Export Audited Manifest to File):** Writes formatted transfer entries and batch summary metrics to `remittance_manifest_output.txt` on disk. Alerts the operator with the count of exported records.
- **Option 5 (Exit Gateway Terminal):** Displays an operational closing notice and terminates the execution loop cleanly via `break` or loop flag.
- **Input Validation:** If the operator submits an out-of-range option or invalid format, display an error message and reprompt the menu without crashing.

Console Display Specifications (Option 3 Sample)

When Option 3 is executed, the application prints a formatted manifest table with aligned columns:

```
=================================================================================================================
                                    ACTIVE SETTLEMENT MANIFEST
=================================================================================================================
Txn ID    Sender Name        Curr  Gross (PHP)      Total Fee (PHP)  Net Payout (PHP)  Compliance Status
-----------------------------------------------------------------------------------------------------------------
TXN-3001  Alice Smith        USD     58,200.00            1,173.00         57,027.00  APPROVED FOR SETTLEMENT
TXN-3002  Victor Markov      EUR     93,750.00                0.00              0.00  SETTLEMENT REJECTED
TXN-3003  Chen Wei           JPY      7,600.00              180.00          7,420.00  HELD FOR COMPLIANCE REVIEW
TXN-3004  Elena Rostova      CAD    126,300.00            1,563.00        124,737.00  APPROVED FOR SETTLEMENT
-----------------------------------------------------------------------------------------------------------------
COMPLIANCE & SETTLEMENT ANALYTICS:
  Total Processed    : 4 transactions
  Approved Transfers : 2
  Held for Compliance: 1
  Rejected Transfers : 1
  Total Gross Volume : PHP 192,100.00
  Total Gateway Fees : PHP 2,916.00
  Average Transfer   : PHP 48,025.00
=================================================================================================================
```

File Storage & Archival Specifications

1. Input Batch File (`remittance_batch_input.txt`)

CSV structure:

`TxnId,Sender,Curr,Amount,Corridor,Purpose,Kyc,Sanction,Risk,Channel,Express`

```
TXN-3001,Alice Smith,USD,1000.00,1,Family Support,true,false,25,Bank Deposit,true
TXN-3002,Victor Markov,EUR,1500.00,2,Commercial Import,true,true,92,Cash Pickup,false
TXN-3003,Chen Wei,JPY,20000.00,1,Tuition Payment,false,false,45,Mobile Wallet,false
TXN-3004,Elena Rostova,CAD,3000.00,2,Consulting Services,true,false,30,Bank Deposit,false
```

2. Export Manifest File (`remittanc`

Pipe-delimited records generated by Menu

```
AUDITED CROSS-BORDER SETTLEMENT MAN
Exported: 2026-10-23
---------------------------------
TXN-3001 | Alice Smith | USD 1,000.
TXN-3002 | Victor Markov | EUR 1,50
TXN-3003 | Chen Wei | JPY 20,000.00
TXN-3004 | Elena Rostova | CAD 3,00
---------------------------------
SUMMARY: Total=4 | Approved=2 | Hel
```

Verification & Test Scenarios

Txn ID	Input Attributes	Calculated Financials	Compliance Status
TXN-3001	Alice Smith | USD 1,000.00 Corridor 1 | Family Support Kyc: Yes | Sanction: No | Risk: 25 Channel: Bank Deposit | Express: Yes	Rate: ₱58.20 | Gross: ₱58,200.00 Comm (1.5%): ₱873.00 | Chnl: ₱50.00 Express: ₱250.00 | Fee: ₱1,173.00 **Net Payout: PHP 57,027.00**	APPROVED FOR SETTLEMENT
TXN-3002	Victor Markov | EUR 1,500.00 Corridor 2 | Commercial Import Kyc: Yes | Sanction: Yes | Risk: 92 Channel: Cash Pickup | Express: No	Rate: ₱62.50 | Gross: ₱93,750.00 *Sanction violation + Risk Score > 85* Total Fee: ₱0.00 (Cancelled) **Net Payout: PHP 0.00**	SETTLEMENT REJECTED
TXN-3003	Chen Wei | JPY 20,000.00 Corridor 1 | Tuition Payment Kyc: No | Sanction: No | Risk: 45 Channel: Mobile Wallet | Express: No	Rate: ₱0.38 | Gross: ₱7,600.00 Comm (Flat): ₱150.00 | Chnl: ₱30.00 Express: ₱0.00 | Fee: ₱180.00 **Net Payout: PHP 7,420.00**	HELD FOR COMPLIANCE REVIEW
TXN-3004	Elena Rostova | CAD 3,000.00 Corridor 2 | Consulting Services Kyc: Yes | Sanction: No | Risk: 30 Channel: Bank Deposit | Express: No	Rate: ₱42.10 | Gross: ₱126,300.00 Comm (1.0%): ₱1,263.00 | Chnl: ₱50.00 Express: ₱0.00 | Fee: ₱1,563.00 **Net Payout: PHP 124,737.00**	APPROVED FOR SETTLEMENT

Grading & Verification Checklist

- ☐ **Modular Method Architecture:** FX conversion, AML screening, fee computation, and batch aggregation are partitioned into dedicated static methods returning named C# tuples. No monolithic `Main` method.
- ☐ **Defensive Parsing:** Ingests external CSV rows, verifies token boundary (`parts.Length >= 10`), and converts numeric and boolean values safely using `TryParse` with `out` parameters.
- ☐ **Loop Mastery:** Continuous console session powered by `do-while` / `while` ; records traversed with `for` / `foreach` ; corrupt rows skipped with `continue` ; session exits cleanly with `break` .
- ☐ **Collection Synchronization:** Multi-column batch data is dynamically accumulated in parallel generic `List<T>` collections without fixed array size overflows.
- ☐ **Deterministic Resource Management:** File readers and writers utilize `using` statements or static `File` helper methods to guarantee file handles close immediately.
- ☐ **Logical Namespace Separation:** Domain calculation logic lives under `FinTechRemittance.Core` , while file and console interfaces live under `FinTechRemittance.IO` .
- ☐ **Zero Compilation Warnings:** Solution compiles cleanly under .NET 8 / 9 with 0 errors and 0 warnings.